

## Practical Description

### Practical No 01

#### PCA on Wine Dataset (Oral Notes)

##### Aim:

To reduce the dimensions of the Wine dataset and visualize wine categories using PCA.

##### PCA (Principal Component Analysis):

A technique used to reduce the number of features while keeping maximum information.

##### Why PCA?

- Reduces dimensionality
- Removes redundant features
- Makes visualization easier
- Improves processing speed

##### Steps:

1. Load dataset
2. Scale data
3. Apply PCA
4. Select principal components (PC1, PC2)
5. Visualize using scatter plot

**PC1:** Contains maximum variance.

**PC2:** Contains second highest variance.

## Parctical No 02

2. Develop a Ridge and Lasso regression model to predict the number of bike rentals based on weather conditions and time. Dataset: Bike Sharing Dataset (UCI)

### Practical No. 2

#### Ridge and Lasso Regression for Bike Rental Prediction

##### Aim

To develop Ridge and Lasso Regression models to predict the number of bike rentals based on weather conditions and time.

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##### Very Short Theory (Oral)

##### What is Regression?

Regression is used to predict continuous values such as bike rental count.

##### What is Ridge Regression?

Ridge Regression adds **L2 Regularization** to reduce overfitting by shrinking coefficient values.

$$\text{Cost} = \text{RSS} + \lambda \sum w_i^2 \quad \text{Cost} = \text{RSS} + \lambda \sum w_i^2$$

##### What is Lasso Regression?

Lasso Regression adds **L1 Regularization** and can remove unimportant features by making coefficients zero.

$$\text{Cost} = \text{RSS} + \lambda \sum |w_i| \quad \text{Cost} = \text{RSS} + \lambda \sum |w_i|$$

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##### Why Use Ridge and Lasso?

- Prevent overfitting
- Improve prediction accuracy
- Handle large datasets

- Feature selection (Lasso)
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### **Dataset Features**

- Temperature
  - Humidity
  - Weather Condition
  - Season
  - Hour
  - Day
  - Bike Rental Count (Target)
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### **Steps**

1. Load Bike Sharing Dataset.
2. Handle missing values.
3. Split data into training and testing sets.
4. Train Ridge Regression model.
5. Train Lasso Regression model.
6. Predict bike rental count.
7. Compare model performance.

## **Practical No 03**

3. Create a multiclass classification model to predict wine quality based on chemical properties. Dataset: Wine Quality Dataset (UCI)

### **Practical No. 3**

#### **Multiclass Classification for Wine Quality Prediction**

##### **Aim**

To create a multiclass classification model to predict wine quality based on chemical properties using the Wine Quality Dataset.

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##### **Very Short Theory (Oral)**

##### **What is Classification?**

Classification is a supervised learning technique used to predict categories or classes.

##### **What is Multiclass Classification?**

When the target variable has more than two classes.

**Example:** Wine Quality = 3, 4, 5, 6, 7, 8

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##### **Input Features**

- Fixed Acidity
- Volatile Acidity
- Citric Acid
- Residual Sugar
- Chlorides
- Free Sulfur Dioxide
- Total Sulfur Dioxide
- Density

- pH
- Sulphates
- Alcohol

### **Target Variable**

- Wine Quality
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### **Steps**

1. Load Wine Quality Dataset.
  2. Preprocess data.
  3. Split into training and testing sets.
  4. Train classification model.
  5. Predict wine quality.
  6. Evaluate model using Accuracy, Precision, Recall, and F1-Score.
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### **Common Algorithms Used**

- Decision Tree
- Random Forest
- Support Vector Machine (SVM)
- K-Nearest Neighbors (KNN)

## Practical No 04

4. Develop a Support Vector Machine (SVM) classification model to predict whether a tumor is malignant or benign based on various features derived from digitized images of fine needle aspirate (FNA) of breast mass. Dataset: Breast Cancer Wisconsin (Diagnostic) Dataset (UCI Machine Learning Repository).

### Practical No. 4

#### Support Vector Machine (SVM) for Breast Cancer Classification

##### Aim

To develop a Support Vector Machine (SVM) classification model to predict whether a tumor is **Malignant** or **Benign** using the Breast Cancer Wisconsin Dataset.

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#### Very Short Theory (Oral)

##### What is SVM?

Support Vector Machine (SVM) is a supervised machine learning algorithm used for classification and regression problems.

##### What is the Goal?

To classify a tumor as:

- **Malignant (Cancerous)**
- **Benign (Non-Cancerous)**

##### What is a Hyperplane?

A decision boundary that separates different classes.

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## **Dataset Features**

Features are extracted from FNA (Fine Needle Aspirate) images of breast masses:

- Radius
- Texture
- Perimeter
- Area
- Smoothness
- Compactness
- Concavity
- Symmetry

## **Target Variable**

- Malignant (M)
- Benign (B)

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## **Steps**

1. Load Breast Cancer Dataset.
  2. Preprocess data.
  3. Split into training and testing sets.
  4. Scale features using StandardScaler.
  5. Train SVM model.
  6. Predict tumor class.
  7. Evaluate accuracy.
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## **Why Use SVM?**

- High accuracy
- Works well with medical data
- Effective in classification problems
- Handles high-dimensional data

## **Practical No 05**

5. Develop a Random Forest classification model to predict the forest cover type based on cartographic variables. Handle missing data and perform feature scaling. Evaluate the model using accuracy, precision, recall, and F1 score. Dataset: Forest CoverType Dataset (UCI).

### **Practical No. 5**

#### **Random Forest Classification for Forest Cover Type Prediction**

##### **Aim**

To develop a Random Forest Classification model to predict the forest cover type using cartographic variables from the Forest CoverType Dataset.

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##### **Very Short Theory (Oral)**

##### **What is Random Forest?**

Random Forest is an ensemble learning algorithm that combines multiple Decision Trees to improve prediction accuracy.

##### **What is Classification?**

Classification is used to predict categories or classes.

##### **What is Forest Cover Type?**

It represents different types of forest land cover such as Spruce/Fir, Lodgepole Pine, Aspen, etc.

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## **Dataset Features**

- Elevation
- Slope
- Aspect
- Horizontal Distance to Water
- Hillshade
- Soil Type
- Wilderness Area

## **Target Variable**

- Forest Cover Type

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## **Steps**

1. Load Forest CoverType Dataset.
2. Handle missing values.
3. Perform feature scaling.
4. Split data into training and testing sets.
5. Train Random Forest Classifier.
6. Predict forest cover type.
7. Evaluate model performance.

## **Practical No 06**

**6. Utilize the K-Means clustering algorithm to segment customers based on their purchasing behavior. Apply the Elbow Method to determine the optimal number of clusters that best represent distinct customer segments for targeted marketing strategies. Dataset: UCI Machine Learning Repository - Wholesale Customers Data Set**

### **Practical No. 6**

#### **K-Means Clustering for Customer Segmentation**

##### **Aim**

**To utilize the K-Means Clustering algorithm to segment customers based on their purchasing behavior and determine the optimal number of clusters using the Elbow Method.**

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##### **Very Short Theory (Oral)**

###### **What is K-Means?**

**K-Means is an unsupervised learning algorithm used to group similar data points into clusters.**

###### **What is Clustering?**

**Clustering is the process of dividing data into groups based on similarity.**

###### **What is Customer Segmentation?**

**Grouping customers with similar purchasing behavior for better marketing strategies.**

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##### **Dataset Features**

- **Fresh**
- **Milk**

- **Grocery**
- **Frozen**
- **Detergents\_Paper**
- **Delicatessen**

## **Output**

### **Customer Groups (Clusters)**

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#### **What is the Elbow Method?**

**The Elbow Method helps find the optimal number of clusters (K).**

- **Run K-Means for different K values.**
  - **Calculate WCSS (Within Cluster Sum of Squares).**
  - **Plot K vs WCSS.**
  - **The "elbow point" indicates the best K value.**
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#### **Steps**

- 1. Load Wholesale Customers Dataset.**
  - 2. Preprocess and scale data.**
  - 3. Apply K-Means clustering.**
  - 4. Use Elbow Method to find optimal K.**
  - 5. Train K-Means with optimal K.**
  - 6. Visualize customer clusters.**
  - 7. Analyze customer segments.**
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#### **Why Use K-Means?**

- **Simple and fast**

- **Easy to implement**
- **Helps in customer segmentation**
- **Useful for targeted marketing**

## **Practical no 07**

7. Use K-Medoids clustering to segment wholesale customers based on their annual spending on different product categories. Identify the optimal number of clusters to effectively group customers with similar purchasing behaviours. Dataset: Wholesale Customers Dataset (UCI).

### **Practical No. 7**

#### **K-Medoids Clustering for Wholesale Customer Segmentation**

##### **Aim**

To use the K-Medoids clustering algorithm to segment wholesale customers based on their annual spending on different product categories and identify the optimal number of clusters.

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##### **Very Short Theory (Oral)**

###### **What is K-Medoids?**

K-Medoids is an unsupervised clustering algorithm similar to K-Means, but it uses actual data points (medoids) as cluster centers.

###### **What is a Medoid?**

A medoid is the most representative data point in a cluster.

###### **What is Clustering?**

Grouping similar customers based on purchasing behavior.

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## **Dataset Features**

- Fresh
- Milk
- Grocery
- Frozen
- Detergents\_Paper
- Delicatessen

## **Output**

Customer Segments (Clusters)

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## **Why K-Medoids?**

- Less sensitive to outliers
- Uses real data points as centers
- Produces stable clusters
- Better for noisy datasets

## **Practical No 08**

8. Develop a Gradient Boosting regression model to predict wine quality based on physicochemical properties. Handle missing values, perform feature scaling, and evaluate the model using MSE, MAE, RMSE, and R-squared. Dataset: Wine Quality Dataset (UCI).

### **Practical No. 8**

#### **Gradient Boosting Regression for Wine Quality Prediction**

##### **Aim**

To develop a Gradient Boosting Regression model to predict wine quality based on physicochemical properties and evaluate it using MSE, MAE, RMSE, and  $R^2$  Score.

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##### **Very Short Theory (Oral)**

##### **What is Gradient Boosting?**

Gradient Boosting is an ensemble learning algorithm that combines multiple weak decision trees to improve prediction accuracy.

##### **What is Regression?**

Regression is used to predict continuous numerical values.

##### **What is Wine Quality Prediction?**

Predicting the quality score of wine based on its chemical properties.

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##### **Dataset Features**

- Fixed Acidity
- Volatile Acidity
- Citric Acid
- Residual Sugar
- Chlorides

- Free Sulfur Dioxide
- Total Sulfur Dioxide
- Density
- pH
- Sulphates
- Alcohol

### **Target Variable**

- Wine Quality
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### **Steps**

1. Load Wine Quality Dataset.
2. Handle missing values.
3. Perform feature scaling.
4. Split dataset into training and testing sets.
5. Train Gradient Boosting Regressor.
6. Predict wine quality.
7. Evaluate using MSE, MAE, RMSE, and  $R^2$  Score.

